What the trace classification actually implies

Lean proves a conditional bound for an actual Weierstrass curve over any finite field. Put $q = |K|$, $N = \#E(K)$ and $t = q+1-N$. If prime r divides N and $t^2 \in \{0, q, 2q, 3q, 4q\}$, then $\mathrm{ord}_r(q) \in \{1, 2, 3, 4, 6\}$, hence at most 6. The same theorem proves $\gcd(r, q) = 1$ and existence of an actual curve point of additive order r. The trace classification is an explicit premise. [pcc:unverified]

Exact integer trace case	A power forced to be one modulo prime r
$t^2 = 0$	q^2
$t^2 = q$	q^3
$t^2 = 2q$	q^4
$t^2 = 3q$	q^6
$t^2 = 4q$	q^2 (indeed $q = 1$ modulo r)

These are symbolic implications, not empirical field samples. Divisibility of the actual point count gives $q+1 = t$ modulo r. The five polynomial identities and a small divisor table yield the listed orders. [pcc:unverified]

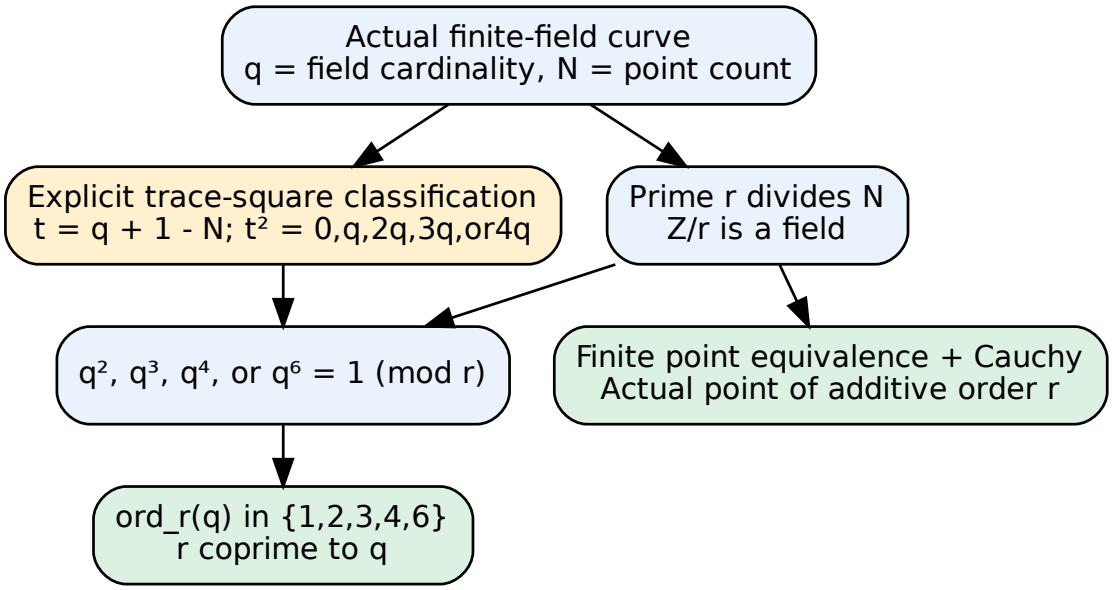


Figure 1: Proof obligations: the trace classification stays an explicit assumption. Actual point existence is obtained separately by Cauchy's theorem.

A negative control that stays within its scope

The combined Lean certificate also checks the arithmetic instance $q = 289$, $N = 324$, $r = 81$. Here $r \mid N$, $\gcd(r, q) = 1$, $t = -34$ and $t^2 = 4q$, while $\mathrm{ord}_{81}(289) = 9 > 6$. Thus primality cannot simply be dropped from the count-only argument. No actual curve or point of order 81 is asserted by this control. [pcc:unverified]

The actual prime-order point theorem is different: the affine point equivalence gives a finite point type, and Mathlib's additive Cauchy theorem supplies a point from prime divisibility of its group cardinality. Local classical decidability supplies the group-law instances. The main theorem quantifies the actual field, curve and prime divisor rather than assuming a modular-order conclusion. [pcc:unverified]

The main binding is `trace_classification_audit_certificate` in SupersingularTraceClassesAudit. The proof map is `../../SUPERSINGULAR_TRACE.md`; exact normalized declaration and source hashes are in `../../supersingular-trace-sources.json`. Lean4.35.0 and Mathlib `56a00585` are pinned. Attempt06 compiles using only propext, Classical.choice and Quot.sound; failed attempts01/04/05 are retained. No custom axiom, hole, native decision exception or linter waiver. [pcc:unverified]

The whole-project gate passes331declarations with zero failures and zero trusted-native exceptions. Source commit `c60c141e` precedes packet `PCC-20261110-9c7d48`; its actual local-check record accompanies this report. Producer-local checks do not create signed verification standing. [pcc:unverified]

Freeman–Scott–Teske (2010), Definition2.1/Remark2.2 pp230–231 and the trace-square list p237 were read. Original Waterhouse Theorem4.1and proof beginning pp536–537were also read and visually inspected after a successful resumed fetch. Its parity/characteristic/congruence restrictions give the necessary square-list premise. Exact read scopes and PDF hashes are in the source plan. No novelty claim or proof of the classification is inferred. [pcc:unverified]

Geometric supersingularity implies the classification, and modular order equals the actual cyclotomic embedding-field degree: these bridges remain separate proof obligations. The older actual-field $r = 2$ disproof remains unchanged. This conditional bound and its numerical control do not finish the general supersingular theorem or the fourteen-item audit. No new native experiment, timing or paid compute. [pcc:unverified]